Static Gesture Classification — Run Report

Generated 2026-05-13 11:37:28

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic · 36 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic · 24 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic · 35 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic · 36 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic · 35 trials
Total training paths	10 · 341 trials total
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Accel
Sensor columns	78 channels
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4212
Split / seed / CV	random_state=42 · CV: LOUO (10 folds, every user held out once)
Sample counts	train (post-aug)=341 · LOUO-evaluated samples=341

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — LOUO cross-validation

Algorithm	CV mean (LOUO, 10 folds)	CV std	CV min	CV max
SVM (Linear)	0.9004	0.0737	0.7429	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	0.8988	0.0715	0.7714	0.9722
Logistic Regression	0.8848	0.0766	0.7429	0.9583
LDA	0.8795	0.0477	0.8000	0.9714
Stacking (SVM+RF+KNN+LogReg)	0.8778	0.0693	0.7429	0.9444
SVM (RBF)	0.8777	0.0681	0.7429	0.9722
Random Forest	0.8704	0.0998	0.6857	1.0000
KNN	0.8254	0.0951	0.6857	0.9583

Per-class CV F1 (across all LOUO folds)

Class	SVM (Linear)	Voting (soft)	Logistic Regression	LDA	Stacking	SVM (RBF)	Random Forest	KNN	Support
Double_Lower	0.962	0.913	0.932	0.952	0.922	0.911	0.923	0.833	50
Double_Nothing	0.667	0.674	0.628	0.590	0.640	0.653	0.667	0.413	46
Double_Pistol_Recoil	0.842	0.900	0.845	0.846	0.833	0.837	0.845	0.907	50
Double_Raise	0.942	0.942	0.942	0.942	0.911	0.939	0.900	0.811	50
Double_Wiggle	0.941	0.941	0.923	0.876	0.969	0.949	0.929	0.867	49
Double_cmere	0.990	0.990	0.990	0.980	0.980	0.980	0.917	0.902	51
Drum_Roll	0.907	0.884	0.874	0.905	0.884	0.871	0.860	0.854	45
macro avg	**0.893**	**0.892**	**0.876**	**0.870**	**0.877**	**0.877**	**0.863**	**0.798**	**341**

Per-user accuracy (each user as held-out fold)

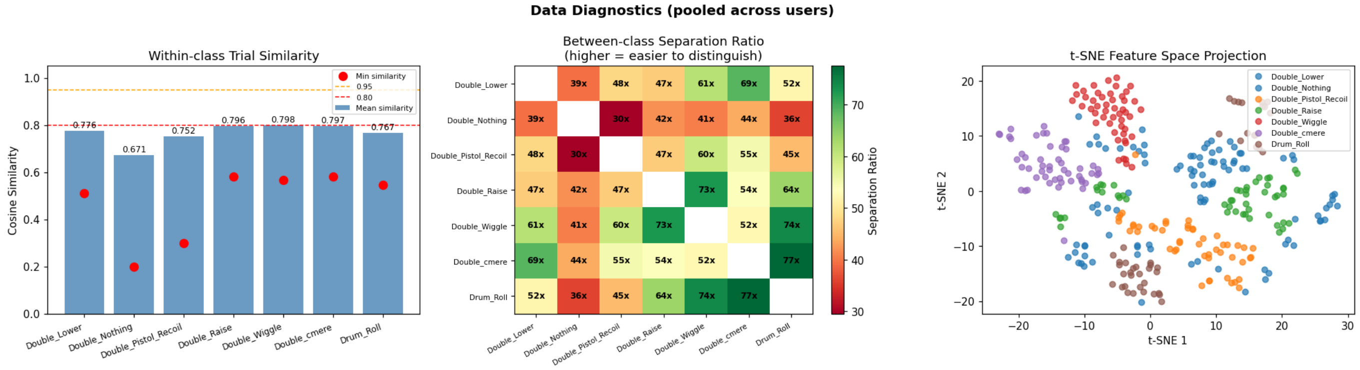
User	n trials	SVM (Linear)	Voting (soft)	Logistic Regression	LDA	Stacking	SVM (RBF)	Random Forest	KNN

1_Alan	35	0.943	0.943	0.943	0.971	0.943	0.914	1.000	0.743
2_Alex	35	0.914	0.914	0.886	0.829	0.800	0.857	0.686	0.686
3_Anghad	36	0.889	0.972	0.917	0.889	0.944	0.917	0.972	0.722
4_Daniel	35	0.914	0.914	0.914	0.886	0.914	0.914	0.886	0.886
6_Harry	35	0.943	0.914	0.943	0.914	0.943	0.914	0.914	0.886
8_Jestin	24	1.000	0.958	0.958	0.875	0.917	0.917	0.958	0.958
11_Mansh	35	0.743	0.771	0.743	0.829	0.743	0.771	0.743	0.857
12_Marcus	35	0.800	0.771	0.743	0.800	0.800	0.743	0.771	0.714
14_StephenV2	36	0.972	0.972	0.944	0.917	0.917	0.972	0.917	0.944
17_Raquel	35	0.886	0.857	0.857	0.886	0.857	0.857	0.857	0.857
Mean (across users)		0.900	0.899	0.885	0.879	0.878	0.878	0.870	0.825
Std (across users)		**0.074**	**0.072**	**0.077**	**0.048**	**0.069**	**0.068**	**0.100**	**0.095**

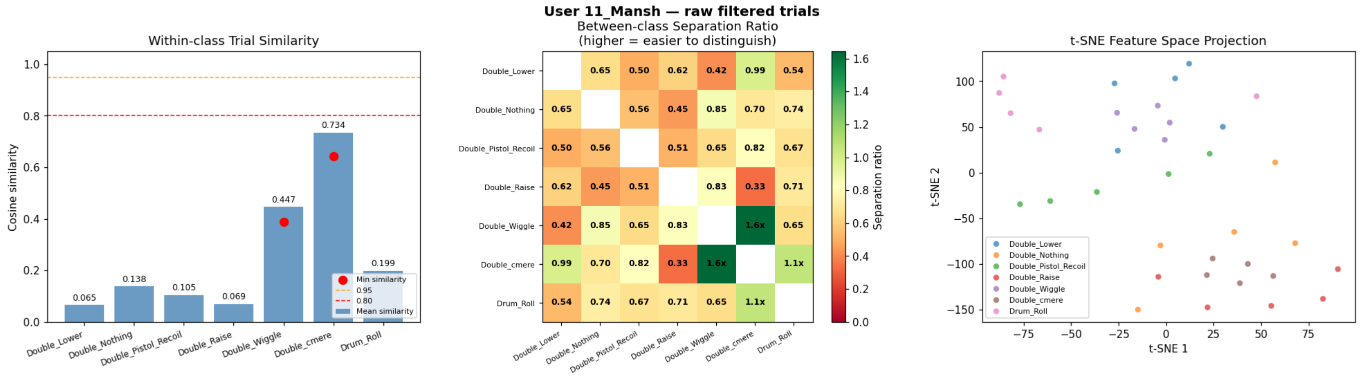
Appendix — Data Diagnostics

42 figures: pooled across users + one per detected user (raw filtered trials and extracted-feature views).

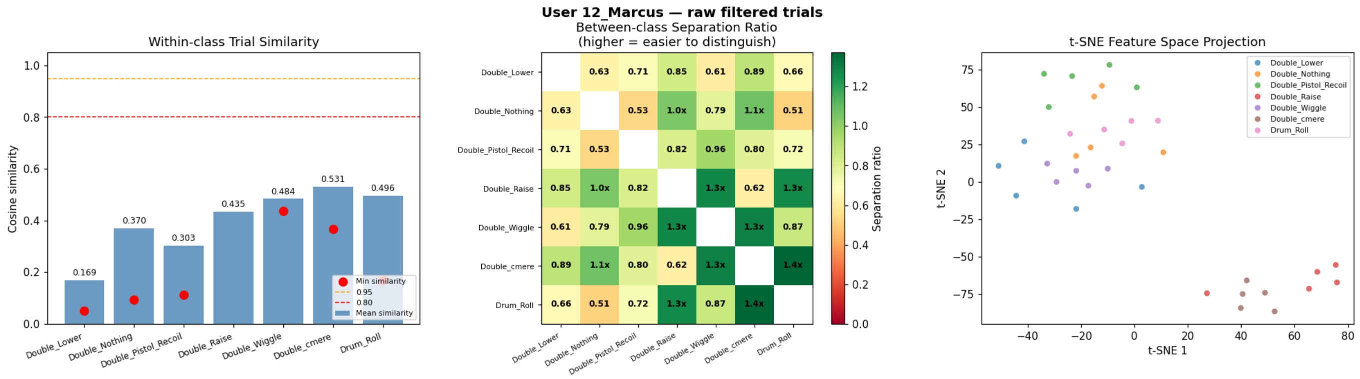
Pooled diagnostics — extracted features (across all users)



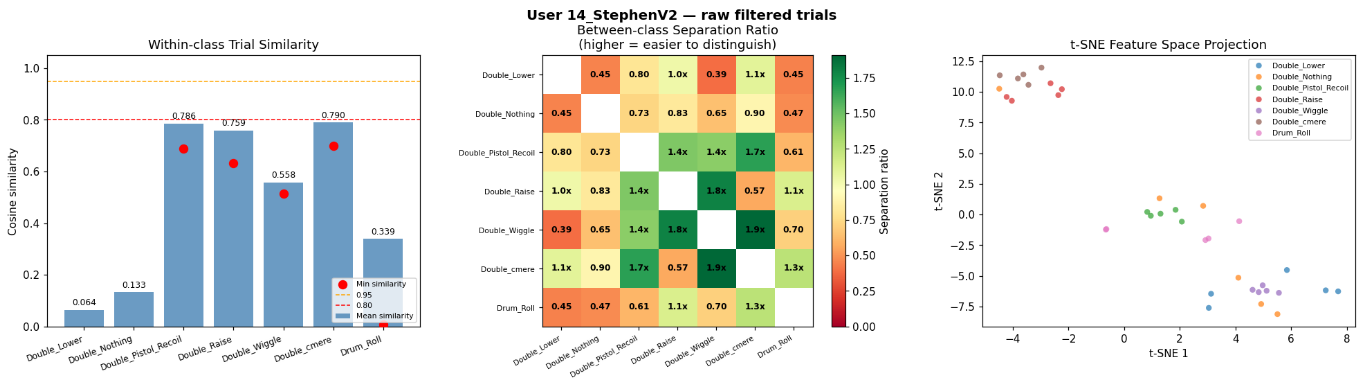
User 11_Mansh — raw filtered trials



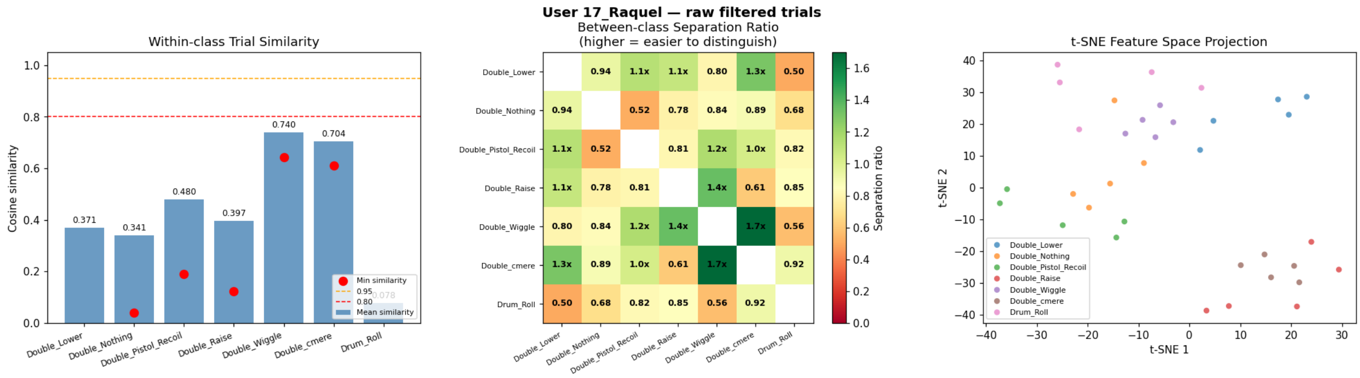
User 12_Marcus — raw filtered trials



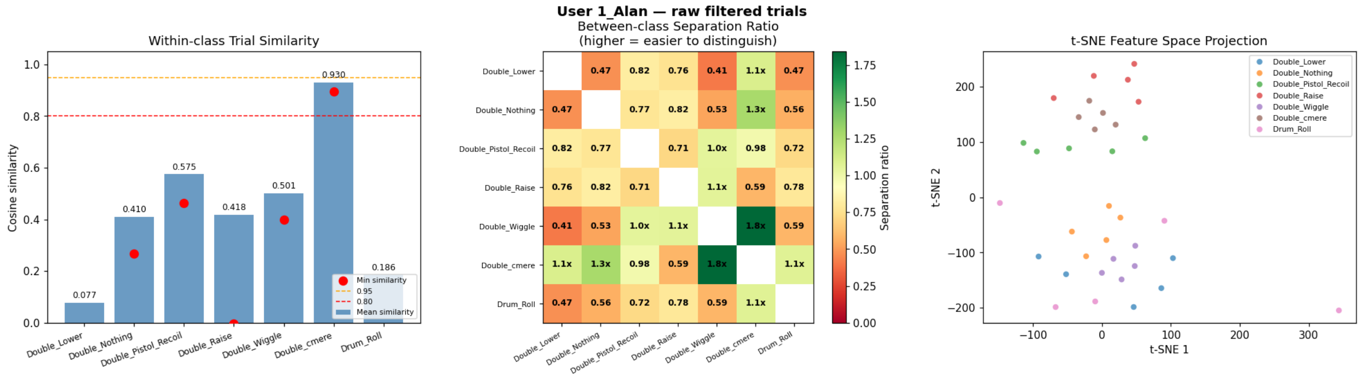
User 14_StephenV2 — raw filtered trials



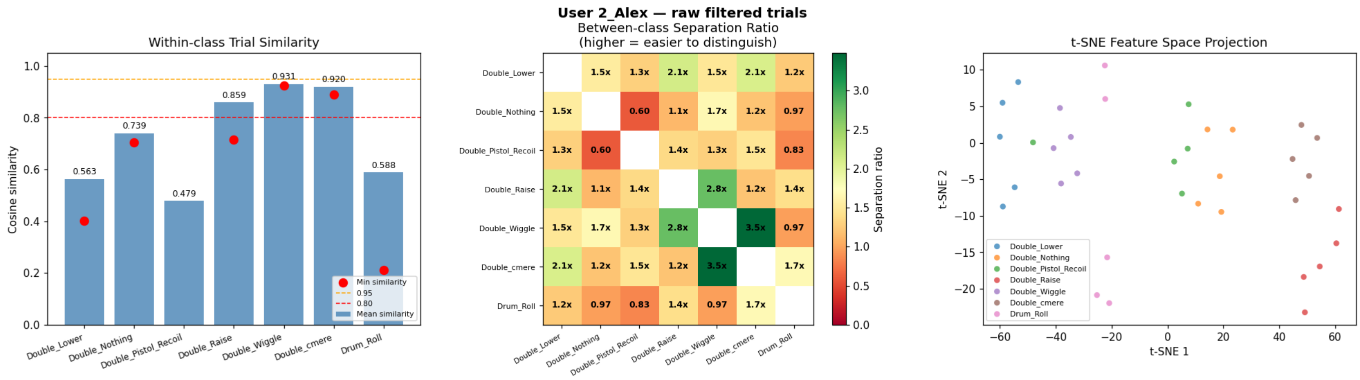
User 17_Raquel — raw filtered trials



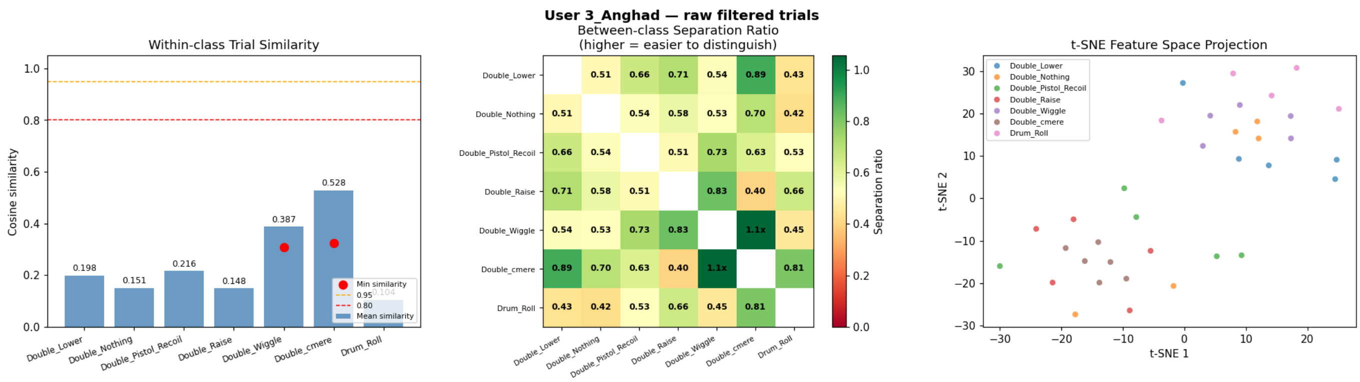
User 1_Alan — raw filtered trials



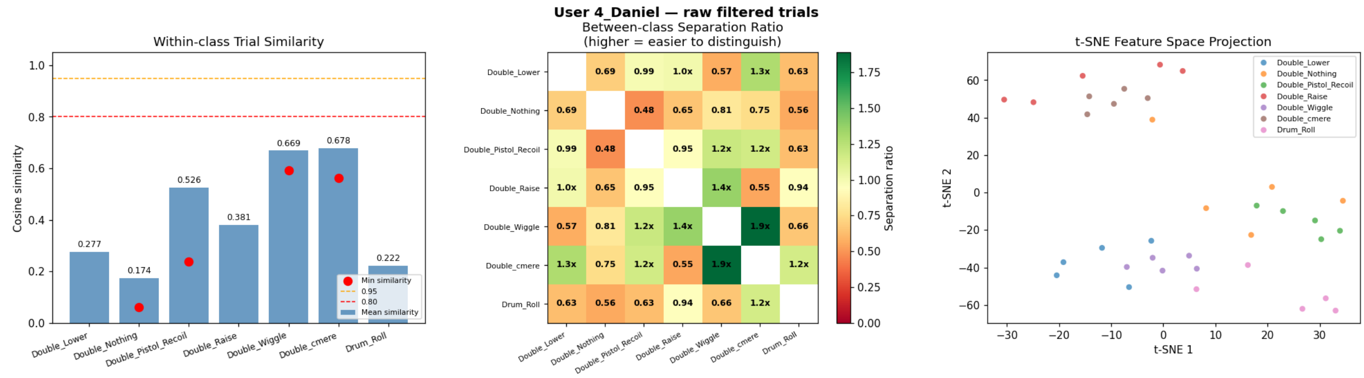
User 2_Alex — raw filtered trials



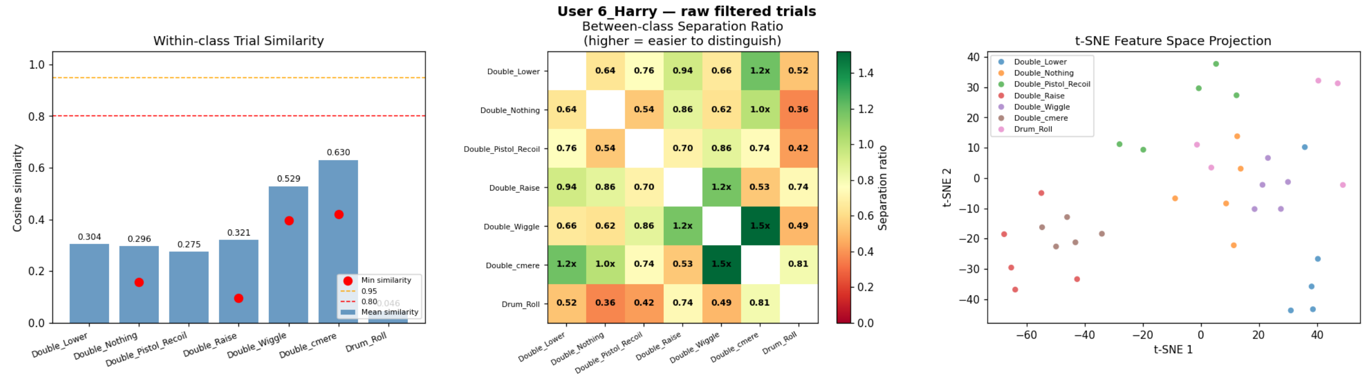
User 3_Anghad — raw filtered trials



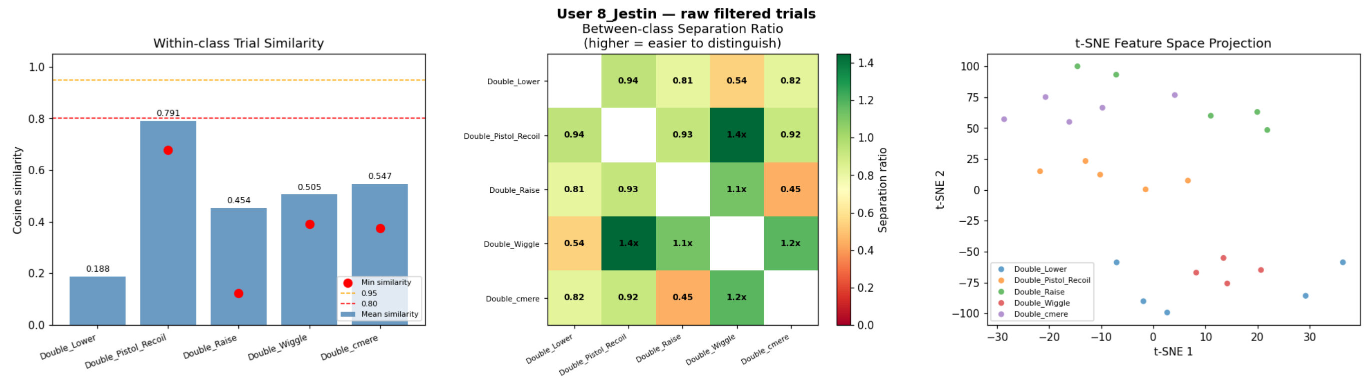
User 4_Daniel — raw filtered trials



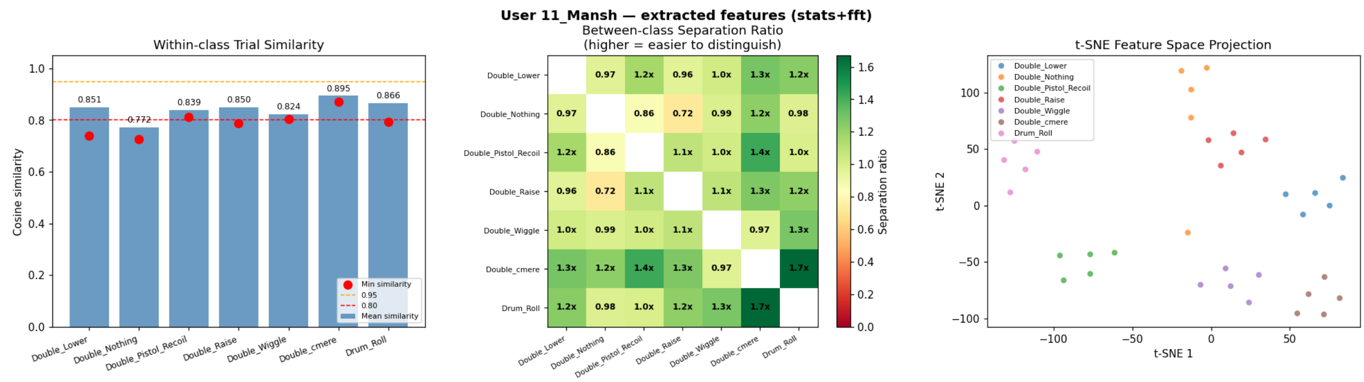
User 6_Harry — raw filtered trials



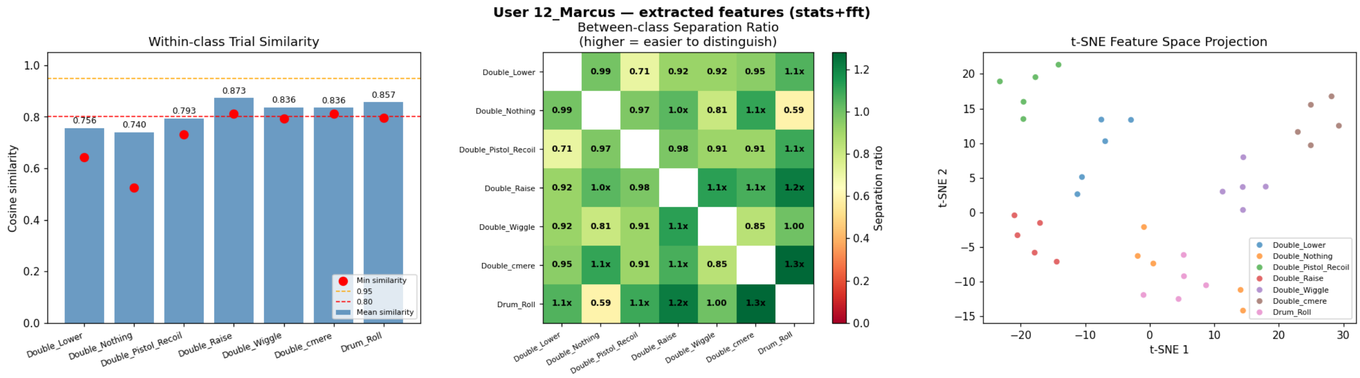
User 8_Jestin — raw filtered trials



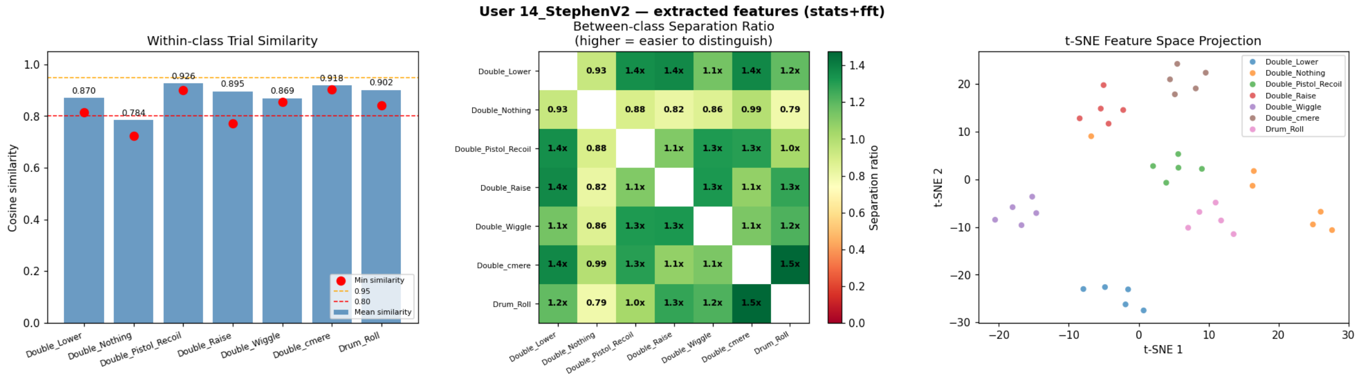
User 11_Mansh — extracted features (stats+fft)



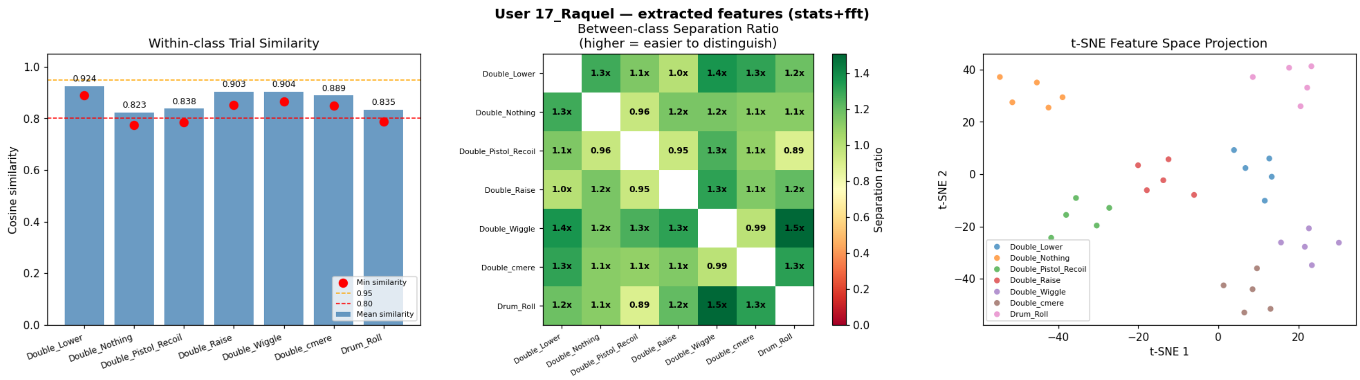
User 12_Marcus — extracted features (stats+fft)



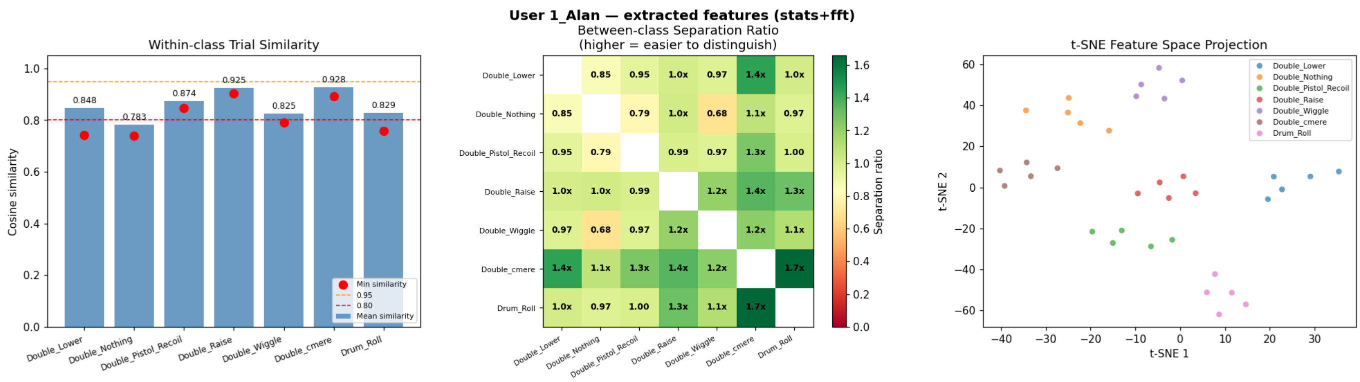
User 14_StephenV2 — extracted features (stats+fft)



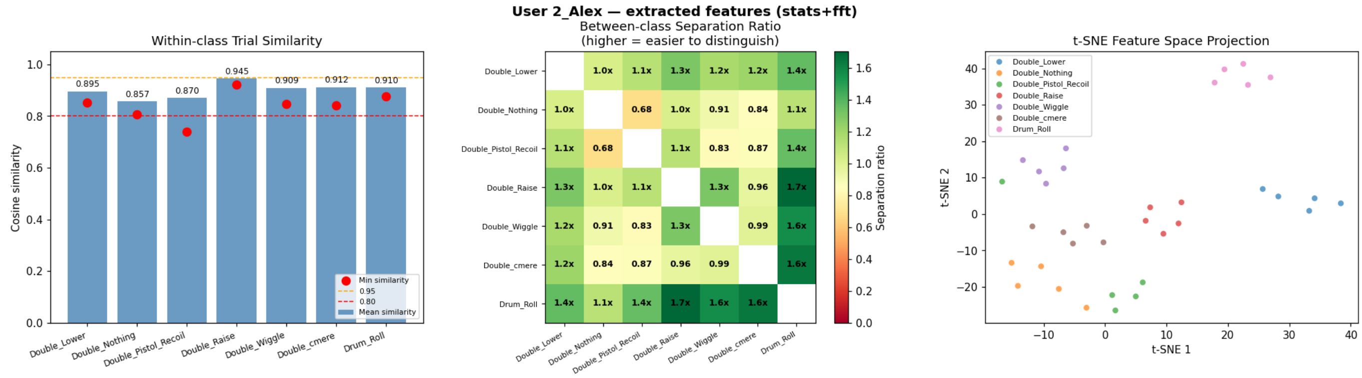
User 17_Raquel — extracted features (stats+fft)



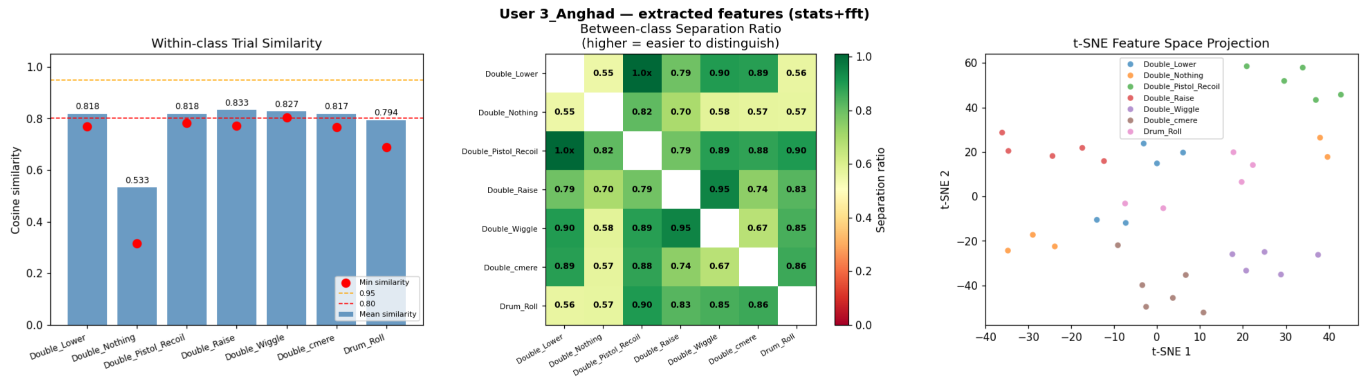
User 1_Alan — extracted features (stats+fft)



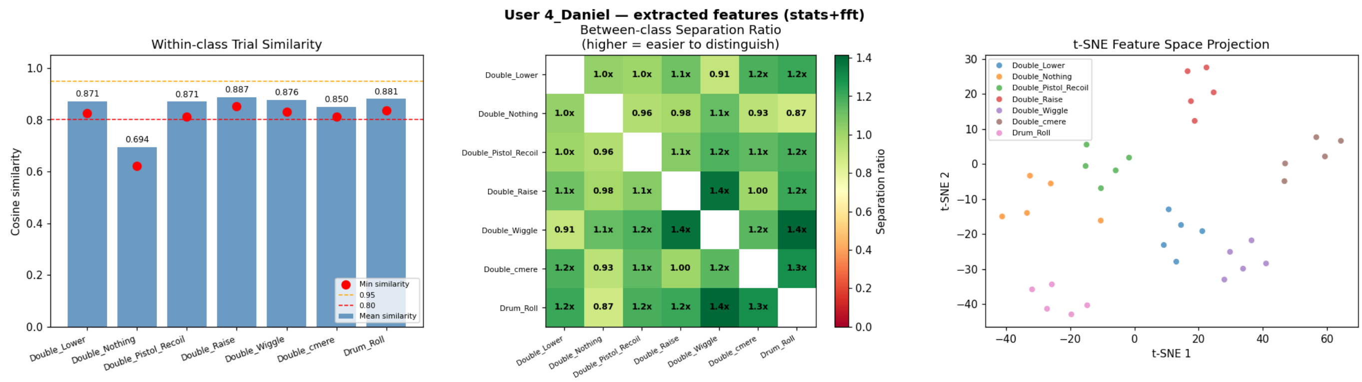
User 2_Alex — extracted features (stats+fft)



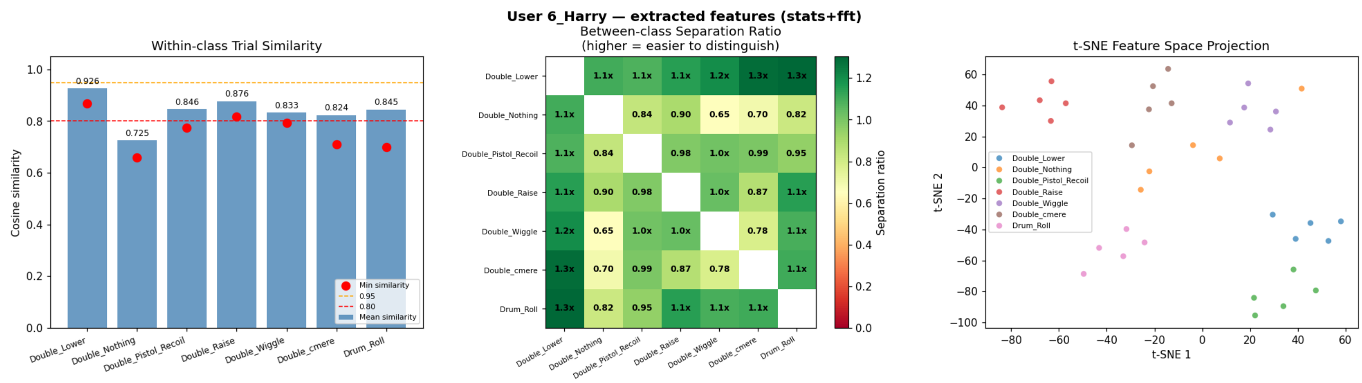
User 3_Anghad — extracted features (stats+fft)



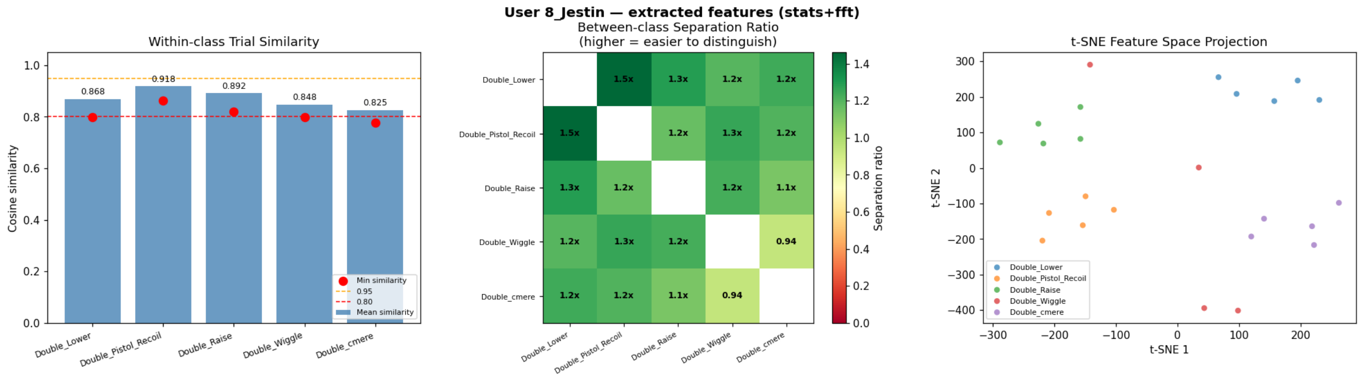
User 4_Daniel — extracted features (stats+fft)



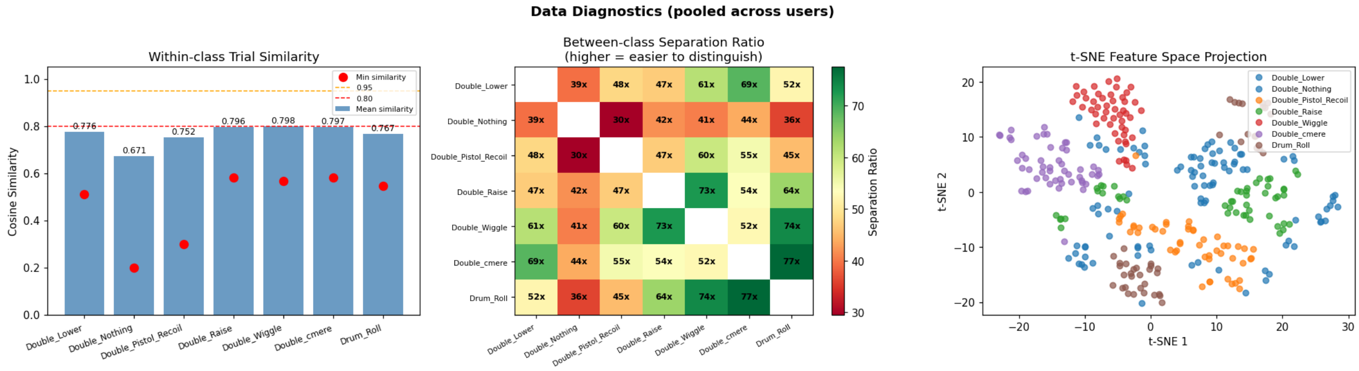
User 6_Harry — extracted features (stats+fft)



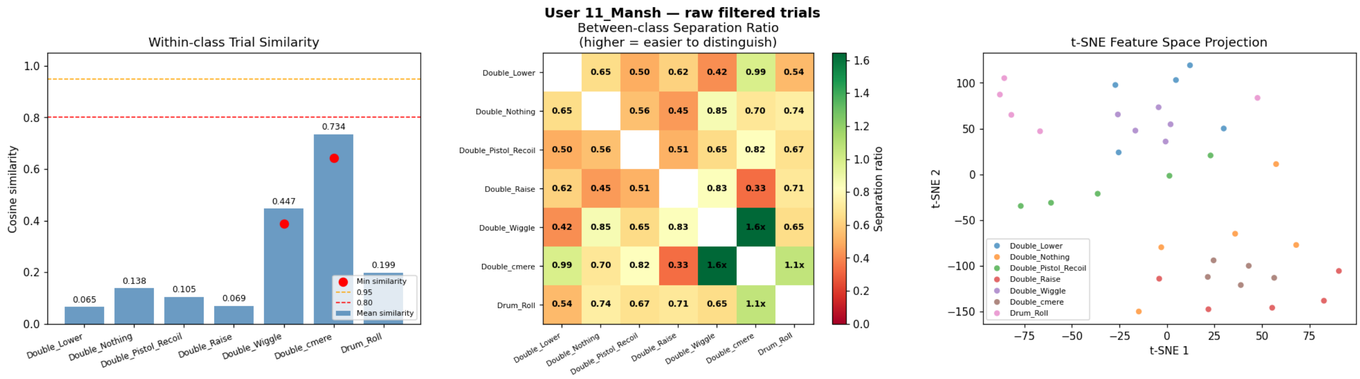
User 8_Jestin — extracted features (stats+fft)



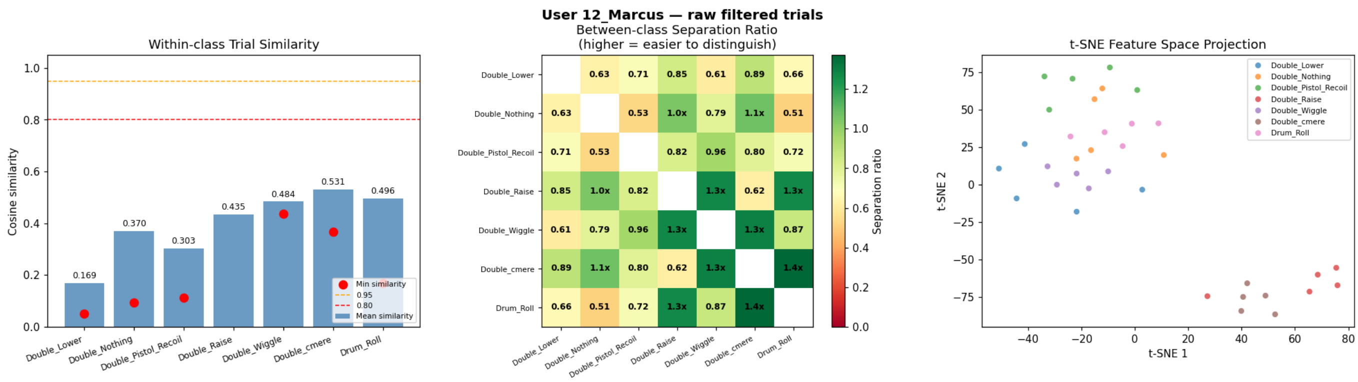
Pooled diagnostics — extracted features (across all users)



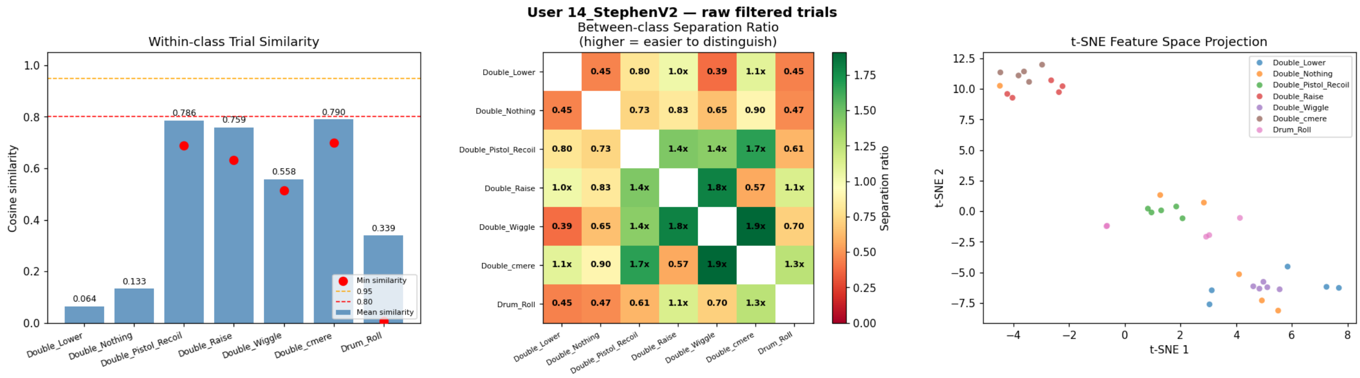
User 11_Mansh — raw filtered trials



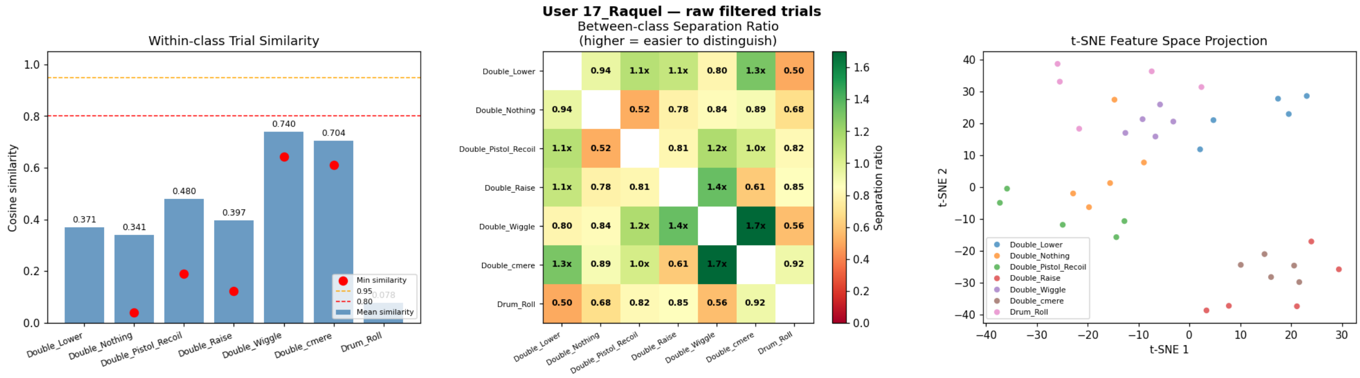
User 12_Marcus — raw filtered trials



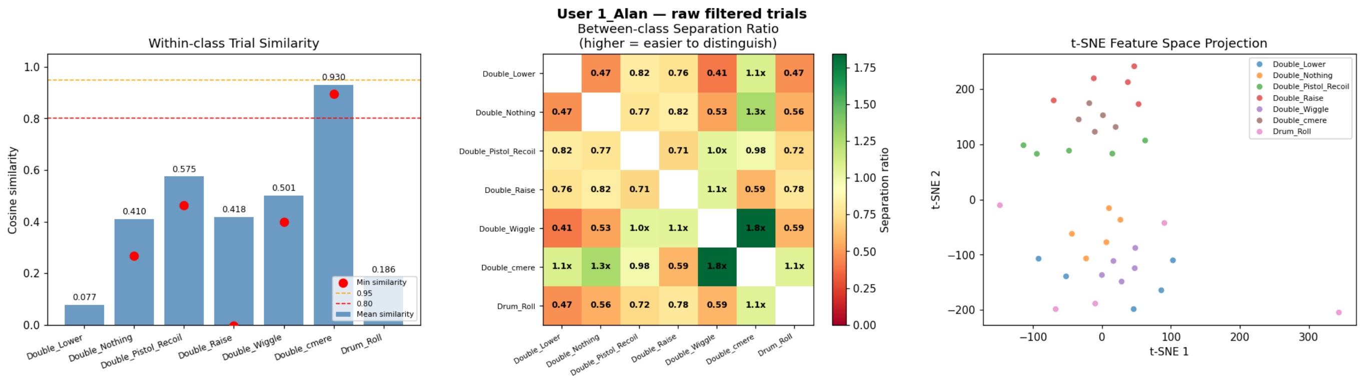
User 14_StephenV2 — raw filtered trials



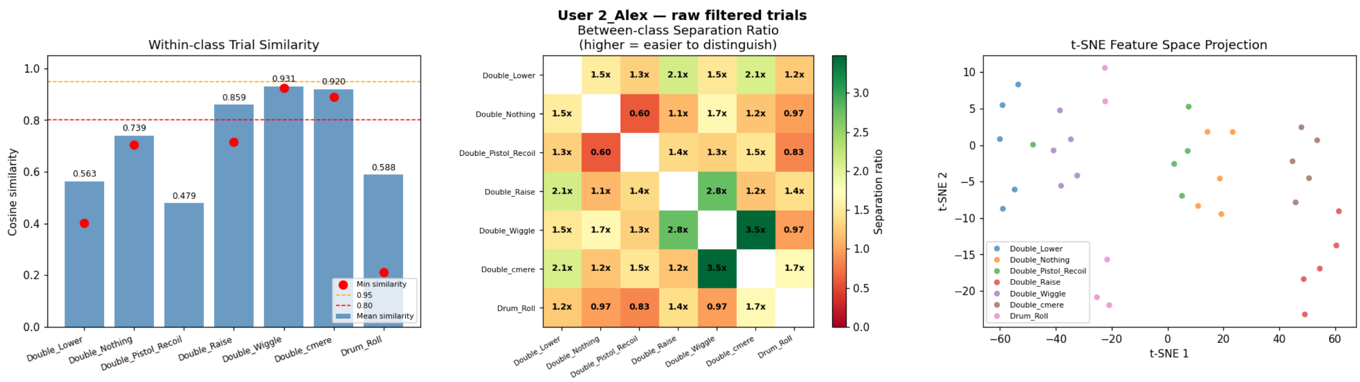
User 17_Raquel — raw filtered trials



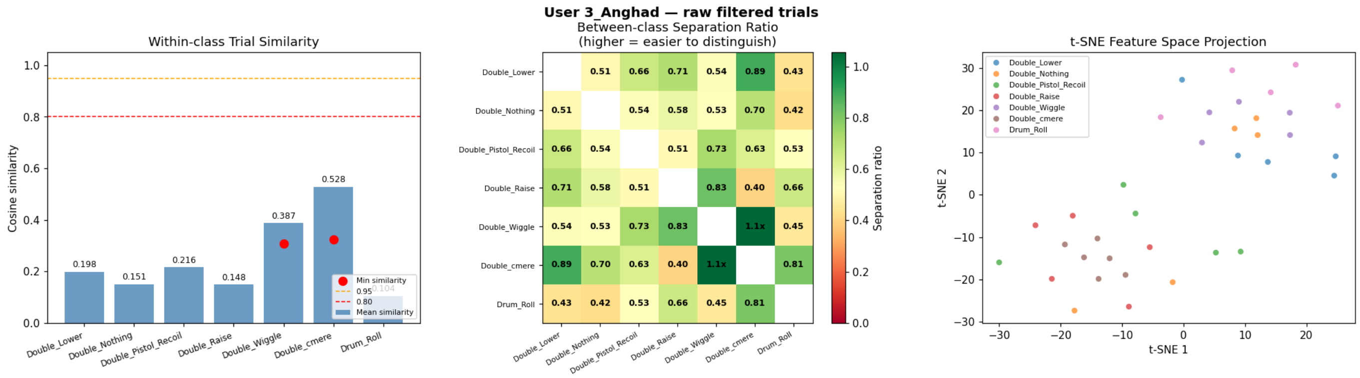
User 1_Alan — raw filtered trials



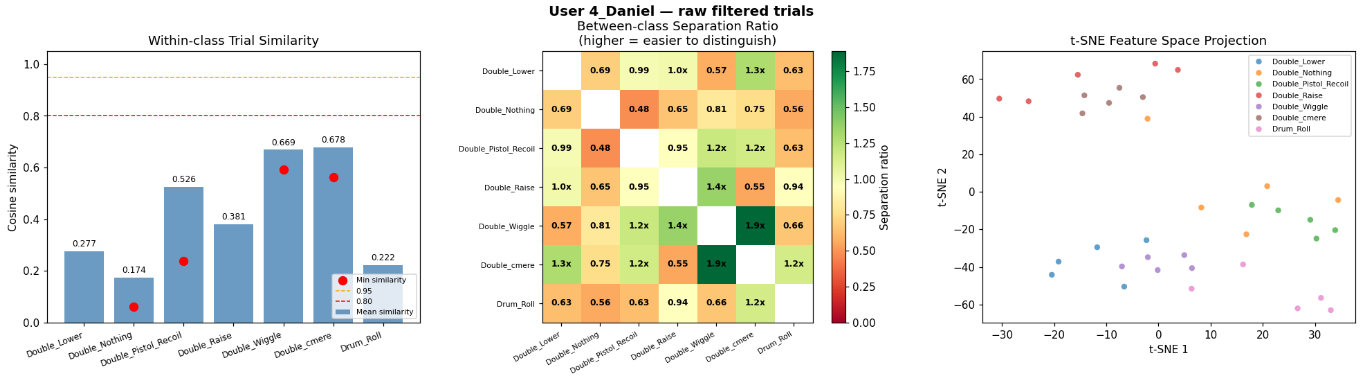
User 2_Alex — raw filtered trials



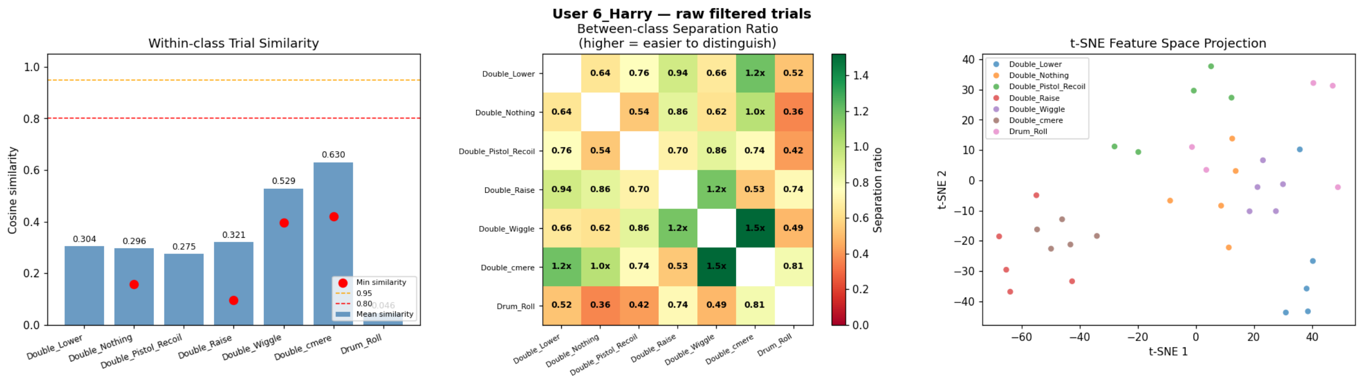
User 3_Anghad — raw filtered trials



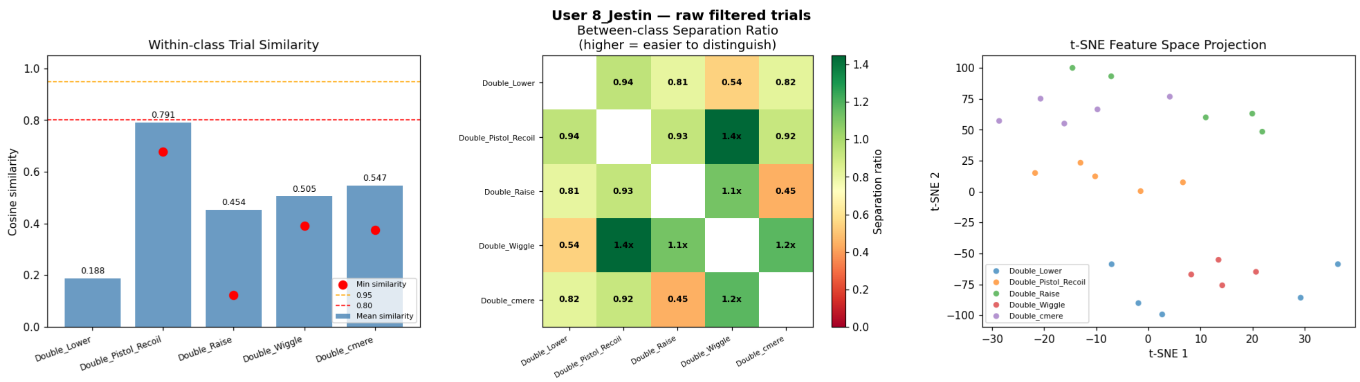
User 4_Daniel — raw filtered trials



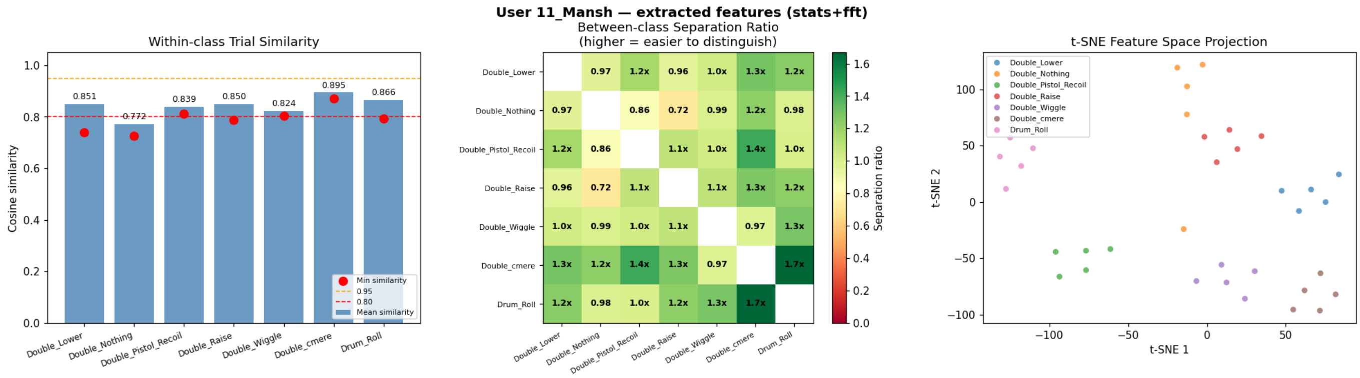
User 6_Harry — raw filtered trials



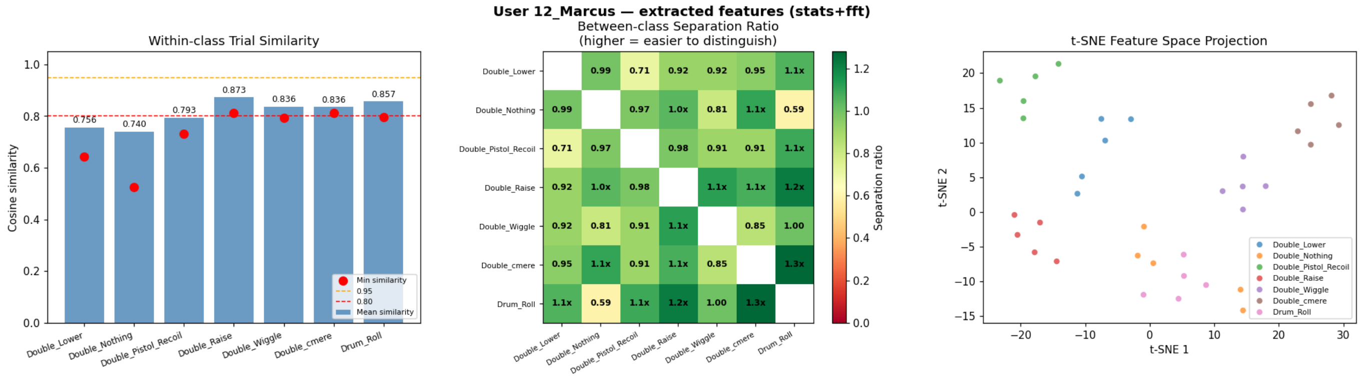
User 8_Jestin — raw filtered trials



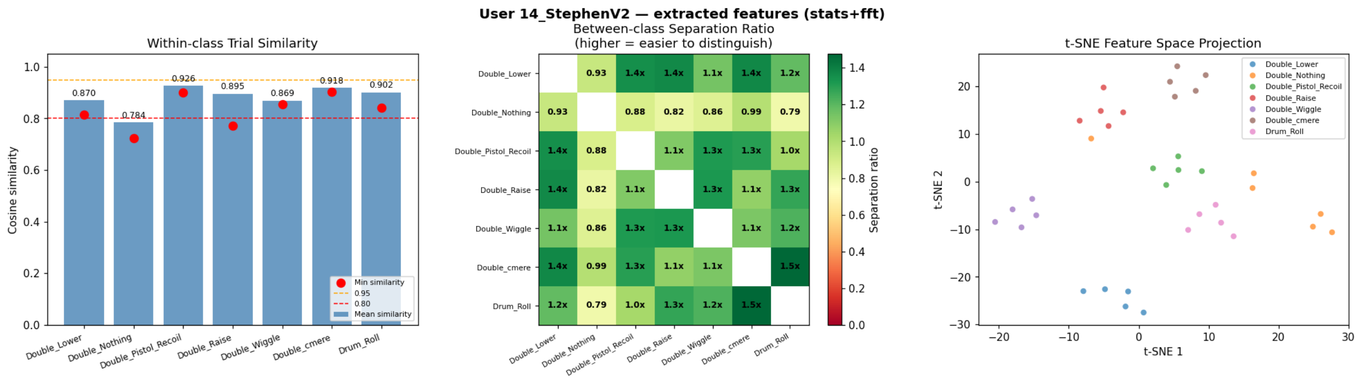
User 11_Mansh — extracted features (stats+fft)



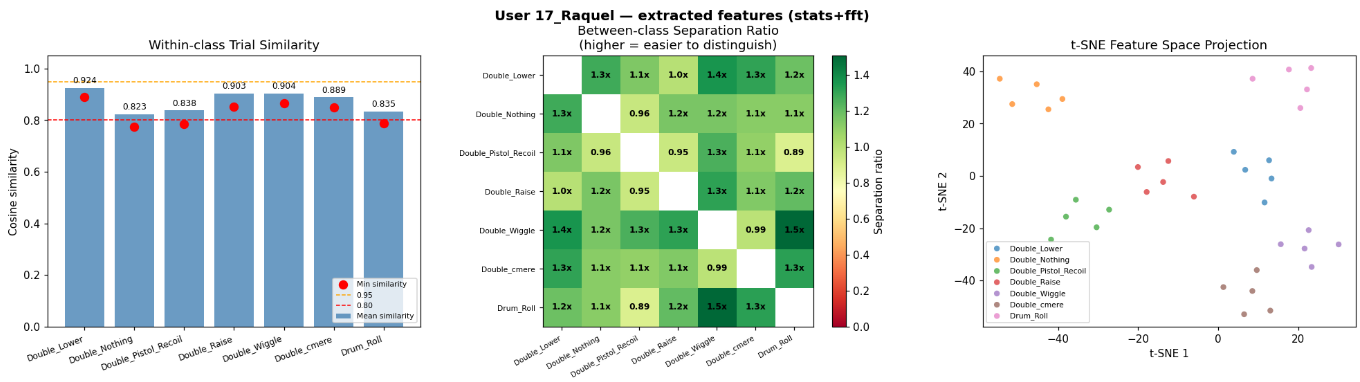
User 12_Marcus — extracted features (stats+fft)



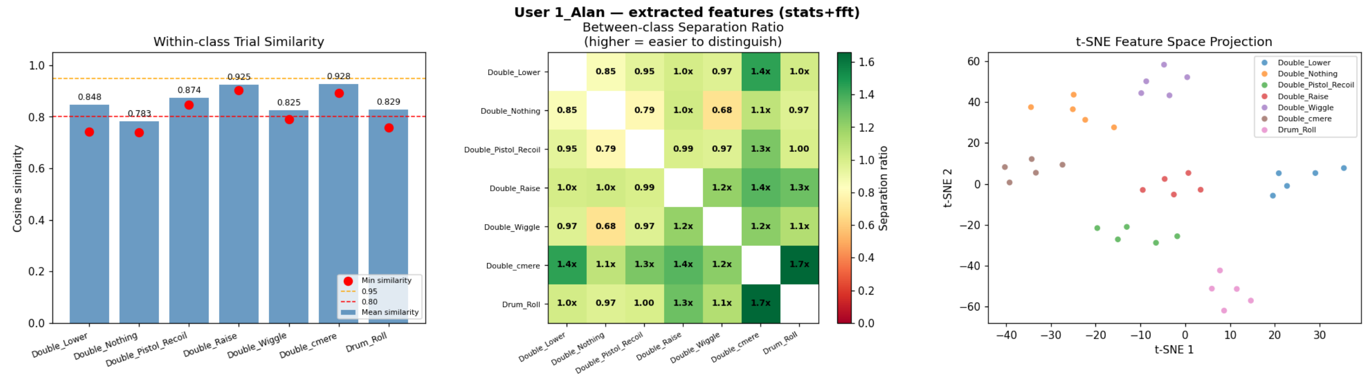
User 14_StephenV2 — extracted features (stats+fft)



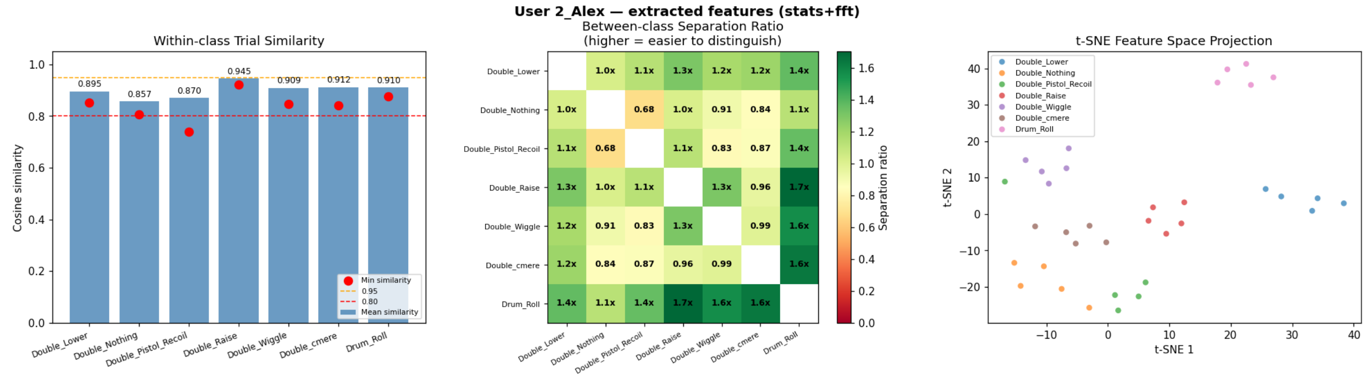
User 17_Raquel — extracted features (stats+fft)



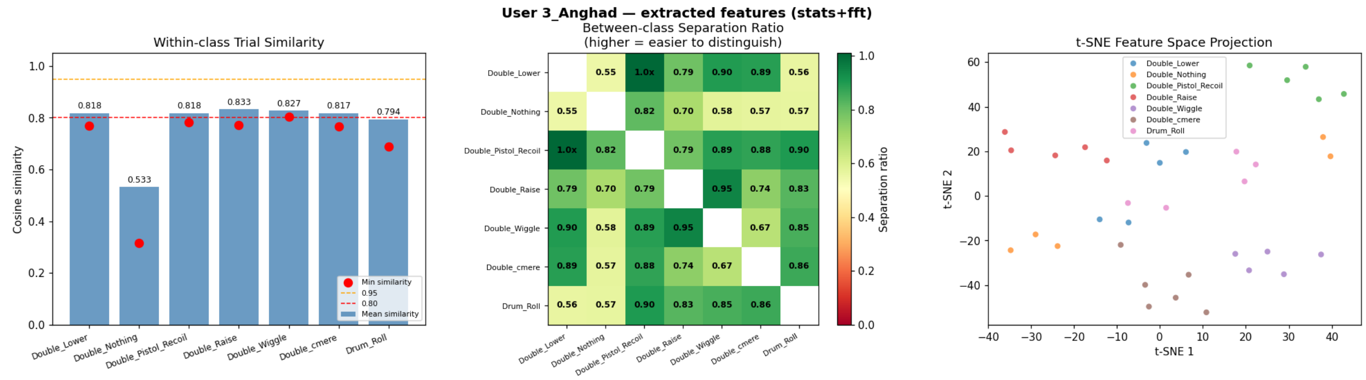
User 1_Alan — extracted features (stats+fft)



User 2_Alex — extracted features (stats+fft)



User 3_Anghad — extracted features (stats+fft)



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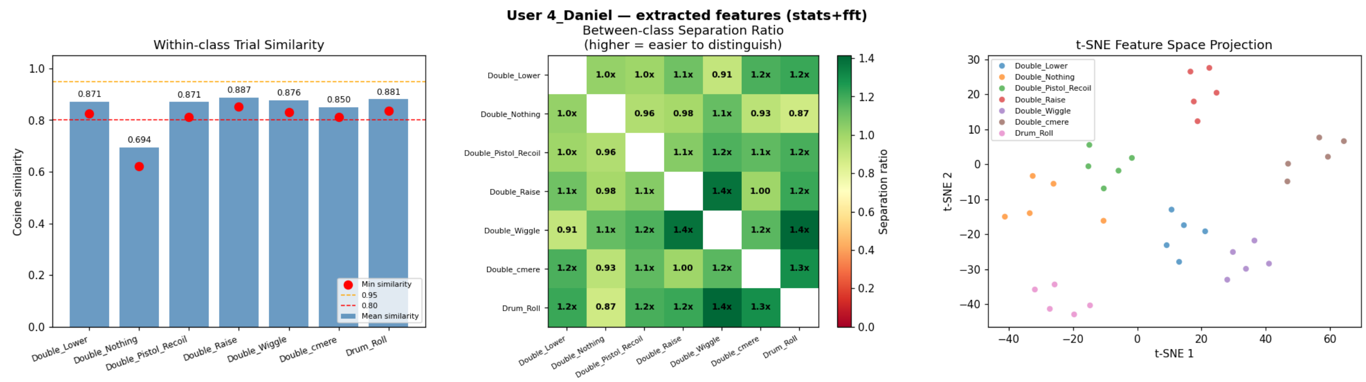

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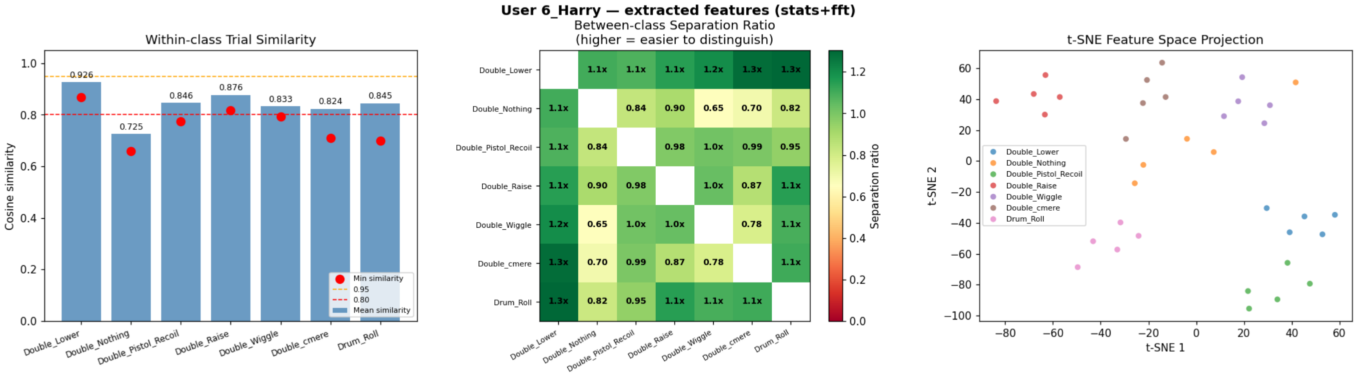

User 8_Jestin — extracted features (stats+fft)